

Design FMEA — t8_r3 (VBF-T8R3-DFMEA-01)

Design FMEA (qualitative version, simulation-signal based) — t8_r3 (VBF-T8R3-DFMEA-01)

Qualitative ratings derived from simulation signal magnitudes vs thresholds; complete process/supplier FMEA is N/A (beyond pure simulation boundary).

	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
cooling	anode_potential_v min +17.7 mV at 4C/45 °C (V13) — margin over 0 V	high	low	negative porosity
rise	T_max 326.53 K vs 333.15 K limit; run-tr triggered=false on overcharge coupling	high	low	forced-air h≥60
	window 4.2 V top (parameter set); no final DFT endorsement run (real_compute=false) — signal NOT verified at molecular level	medium	low	voltage window
at mass cap	ED 497.5 vs 446.18 Wh/kg (51 Wh/kg margin); retention 98.66% vs 90%	medium	low	V13 loading/po
ation at 4C	V9/V11 (neg 100 μm) anode −26.5/−10.8 mV vs +7.3 mV at 85.2 μm	medium	low (avoided)	negative thickn

****Conclusion**:** no high-risk items remain at the selected design point; the highest-severity modes (plating, thermal) both show positive margins (+17.7 mV, −6.6 K); complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary).